Science Education Tutoring

Exercises

1. Which of the following expressions is equivalent to the expression $\frac{(2^{-3})(2^{5})}{(2^{7})}$?
   a. 32
   b. $(2^{-3+5})(2^{-7})$
   c. $\frac{2^{5}}{2^{3-7}}$
   d. $(2^{5})(2^{-4})$

2. Which of the following $-5,\ \frac{1}{2},\ 3,\ 1$ is the root of the following equation:

$$\frac{5x-8}{3} = \frac{1-3x}{2}$$

   a. -5
   b. $\frac{1}{2}$
   c. 3
   d. 1

3. In two water tanks there is a total of 150 litres of water. If 13 litres of water is removed from the first tank and 35 litres from the second tank, then both tanks have the same amount of water left. How many litres of water does the second water tank have?
   a. 86 litres
   b. 98 litres
   c. 75 litres
   d. 100 litres

4. Given $A = -2i + 13j$ and $B = 2(i - 2j)$. Then $A + 2B =$?
   a. $5j + 2i$
   b. $2i - 5j$
   c. $i + 5j$
   d. $2i + j$

5. Ratha, Jay, Seihak, Netta collected 100 eggs from their respective farms. The table below shows the number of eggs they each collected compared to the number of eggs.

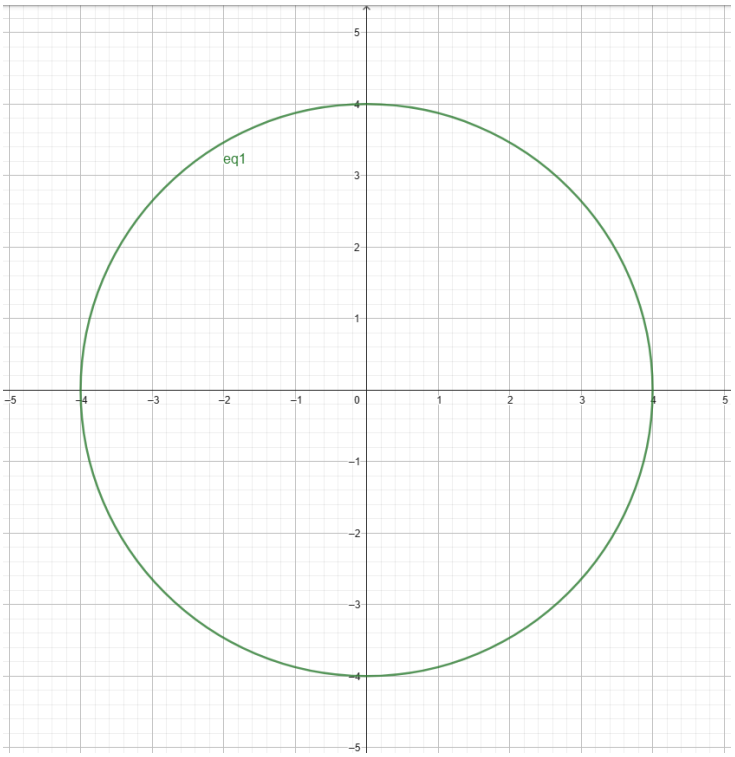
Number of eggs collected	
Names	Number of eggs collected compared to the total number of eggs
Ratha	$\frac{1}{4}$
Jay	0.31
Seihak	26%
Netta	$\frac{9}{50}$

Which one is listed as the one who collects the most eggs and the one who collects the least eggs?

   a. Ratha, Jay, Seihak, Netta
   b. Jay, Seihak, Netta, Ratha
   c. Ratha, Netta, Seihak, Jay
   d. Jay, Seihak, Ratha, Netta

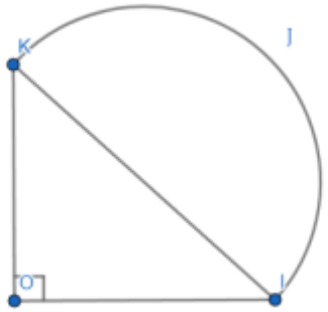
6. The speed of light is 299 792 458 metres per second. Which one is closest to the speed of light?
   a. $3.0 \times 10^8\ m/s$
   b. $2.9 \times 10^6\ m/s$
   c. $2.9 \times 10^8\ m/s$
   d. $3.0 \times 10^6\ m/s$

7. What is the perimeter of the below circle?



   a. 1254 units
   b. 2512 units
   c. 5224 units
   d. 2532 units

8. OIJK is a circular segment whose radius is 25 cm and the central angle is 90°. What is the perimeters of this segment ($\pi = 3.14$)



   a. 89.25 cm
   b. 64.57 cm
   c. 124.12 cm
   d. Other

9. Which expression has the same value of -7?
   a. $|3| + |4|$
   b. $|4| - |3|$
   c. $|-2| - |-9|$
   d. $|9| + |-2|$

10. Chakriya walks down 6 stairs with 10 steps at each stair. Which describes Chakriya walking down the stairs?
   a. $(-6)(-10) = 60$
   b. $-6 + 10 = 4$
   c. $(-6)(10) = -60$
   d. $(-6) + (-10) = -16$

11. Which expression represents 10 less than six times the number n?
    a. $10 - n$
    b. $n - 10$
    c. $10 - 6n$
    d. $6n - 10$

12. The three steps below are used to find the expression equivalent to the expression $\frac{2}{5}(15x - 30y) + 10$

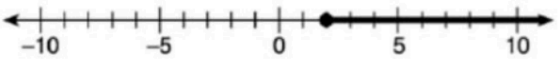
- Step 1: ?
- Step 2: $16x - 12y$
- Step 3: $4(4x - 3y)$
    a. $15(x - 2y) + 10x$
    b. $6x - 30y + 10x$
    c. $6x - 12y + 10x$
    d. $\frac{2}{5}(25x - 30y)$

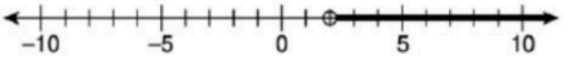
13. What is the sum of expressions $(4a - 5b)$ *and* $(-13b + 6a)$
    a. $10a + 8b$
    b. $10(a - 18b)$
    c. $9a - 18b$
    d. $2(5a - 9b)$

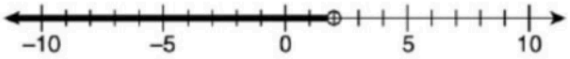
14. Chakriya plays badminton competition at her school and won 16 matches last year. This year Chakriya won 20 matches. What is the percentage increase of the number of matches that Chakriya won from last year to this year?
    a. 20%
    b. 25%
    c. 31%
    d. 80%

15. Which graph best represents the answer of the below inequation?

$$5n + 2 \leq 12$$

a. 

b. 

c. 

d.

16. Chakriya earns i riel, and she spends j riel per month. If the ratio of the amount of income she earns and the amount of money she spends is 8:3. How much does Chakriya earn per month if she saves 75000 riels?
    a. 120000 r
    b. 142000 r
    c. 160000 r
    d. 154000 r

17. What is the value of the coefficient b which satisfies the equation for any real number x?

$$-3(2x-3)+5x=b+9$$

    a. -5
    b. -16
    c. 7
    d. -1

18. A cylindrical water tank with 20 cm height and 320π cm^3 volume. What is the diameter (in metres) of the water tank?
    a. 12
    b. 9
    c. 10
    d. 8

19. What is the answer to the following equation?

$0.2(d - 6) = 0.3d + 5 - 3 + 0.1d$

a. -16
b. 6
c. 12
d. -8

20. Chakriya bought 7 identical key chains and had $4.20 left. If he had bought 12 identical key chains and 4 identical labubu dolls at $0.80 each, she would need $8.50 more. How much did Chakriya have at first?

a. $12.00
b. $24.50
c. $17.5
d. $100